Coral Microbioreactors For Model Validation

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Fig.1: Coral scale

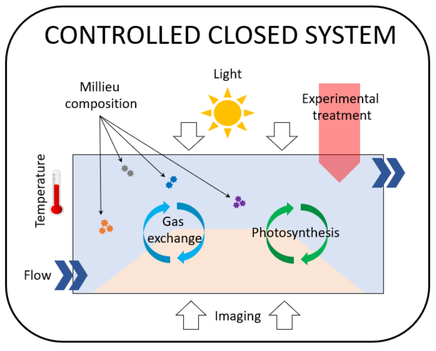
1. Introduction

Advances in coral biology and ecology are hindered by the complexity of the system (symbiotic relationship with dinoflagellate algae, dense and diverse microbiome) and scales at which different mechanisms operate. While colony-level studies are widespread, there are fewer single polyp scale and cellular level studies (Fig.1). Coral research has mainly been conducted top-down by testing colony response to describe cellular processes. Developing coral-specific microbioreactors can help us change to a bottom-up approach that can have the double function of feeding computational models and validating the ensuing predictions. They can, for example, enable us to investigate protein interactions and gene expression applied to the response of corals to environmental change, toxicity (e.g. microplastics), disease and injury at the cellular level, the polyp level and the colony level.

A number of "coral-on-a-chip" designs have been developed to study micro-scale interactions between corals and their environment but they mainly allow investigations at the polyp scale[1-4] or colony scale (nubbin: colony subset)[5,6]. The device by van Treuren et al.[7] presents the highest resolution by trapping individual larvae in a funnel-like channel allowing single-larvae imaging during the experiment (symbiont expulsion during thermal stress-induced bleaching in symbiotic anemone Aiptasia). Despite the technological advances these devices represent, control of environmental properties and number of replicates remain limited, and the cellular scale has still not been achieved.

The design presented here offers multiple independent input/output channels to control growth media or treatment parameters (flow, temperature, composition), light (for photosynthesis) and live-imaging capabilities (traditional or inverted microscopy) Fig.2-3. It also permits multiple configurations (single chamber or network, Fig.3-4) to study biological interactions at different scale with simple logistics.

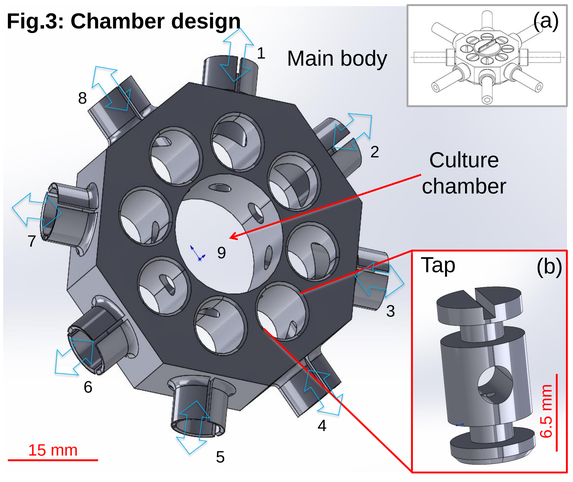
Fig.2: Parameters to control



2. Materials & Methods

Chamber design – Reef-building corals require specific environmental conditions (*e.g.* light, temperature, salinity, pH, nutrient, ammonia). This device was designed to enable control over irradiance (transparent material VeroClear®), temperature (external heat source or heated media), flow (9 input/outputs, Fig.3), media composition, photosynthesis, gas exchange and experimental treatment while allowing observation using microscopy (top and/or bottom) Fig.2-3. This octagonal microfluidic device also offers the potential for single or multiple well configuration (Fig.4, well volume: 3.62 cm^3), which allows investigations at multiple scale (cell<polyp<colony, Fig.1), and can be fitted with microsensors. It is easy to make (3D polyjet printer: Objet Eden 260VS, VCU Innovation Lab), setup and transport with potential for field applications.

Fig.3: Chamber design



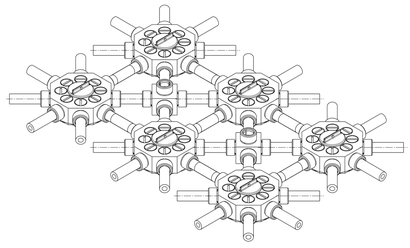
Microfluidic parameters – Shear stress and velocity fields were modeled using computational fluid dynamics simulations of COMSOL Multiphysics® software. The simulations were aimed at optimizing the device design by varying valve diameter, well diameter and valve length. A flow rate of 2 mL/hour was applied and flow was divided equally across inlets when multiple inlets were used. Furthermore, to model the simulations, a laminar, incompressible flow field was assumed due to the low Reynolds number (RE<<2000) and a no-slip boundary condition was applied to the interior of the device to account for viscous effects.

Simulations also focused on the potential chemical transport through the system in relation to chemical dosing and exposure. For this, 180 mg/L of glucose were modeled with the flow rate of 2 mL/hour. Fick's law of diffusion[8] and the mass continuity equation govern the diffusion of the species through time[9].

Live testing:

Planned for January 2021

Fig.4: Network configuration



3. Preliminary Results

Computer simulations were essential to model flow and inlet/outlet configurations considering the many options available. Over all the configurations tested, maximum shear stress and velocity remained low (0.62-2.6x10^{-4} m/s and 0.38-3.1x10^{-4} Pa, respectively) Table 1. The time to steady state solution was 5 hours. Variations in well diameter, valve diameter and valve length (Fig.5) revealed larger diameters (well and valve) and longer valves resulted in lower maximum velocity (0.62 m/s) and lower shear stress (0.38x10^{-4} Pa) Table 1.

Photosynthetically Active Radiation (PAR) measurements (dry, Apogee®) show the device lid reduces light penetration to the well by ~30% compared to the external light (Fig.6).

Of the two external heat sources tested for media temperature control only the hot plate setup was successful at reaching and maintaining the target temperature of 25°C (20min).

Further computer-assisted simulations of flow and chemical transport highlighted the need for intermediate connectors in the network configuration to ensure multiple wells can be dosed simultaneously (four at a time, Fig.7). The configuration including connectors showed no significant increases in flow velocity or shear stress inside the culture wells. The highest shear stress (~1x10^{-4} Pa) and flow velocity (0.7x10^{-4} m/s) were observed in the center of the connectors.

Fig. 5: Device internal geometry and shear stress modelling

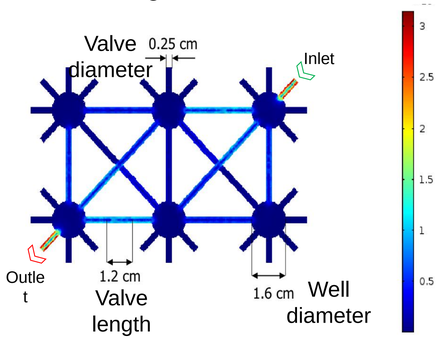


Fig. 6: PAR measurement comparison (dry)

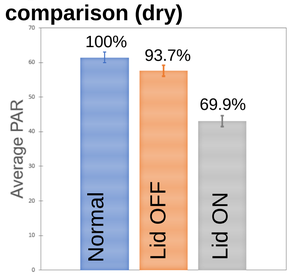


Fig.7: Network with connectors

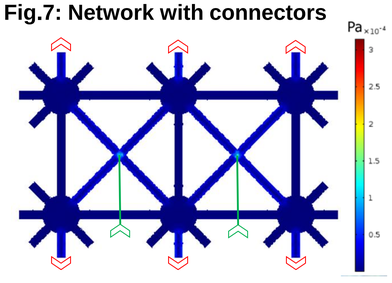


Table 1: Maximum velocity and shear stress (flow rate: 2mL/hour)

	Valve diameter (cm)	Well diameter (cm)	Valve length (cm)	Maximum velocity (m/s)	Maximum shear stress (Pa)
Test 1	0.5	1.6	1.2	**0.62 x10^{-4}**	**0.38 x10^{-4}**
Test 2	0.25	1.6	1.2	2.3 x10^{-4}	2.8 x10^{-4}
Test 3	0.25	1.2	1.2	2.4 x10^{-4}	2.9 x10^{-4}
Test 4	0.25	1.2	0.8	2.6 x10^{-4}	3.1 x10^{-4}

4. Discussion

Although preliminary results are limited and more testing remain the octagon design is promising. We were able to design and tailor a microfluidic device permitting single well and network configuration to study transport and exchanges between multiple "sibling" organisms (or cells). The device's practical aspect means fitting microsensors for continuous measurements will be possible.

Live testing will enable us to check for potential further adjustments, and key parameters/measurements to be uploaded to machine learning platforms currently being developed specifically for the study of reef-building corals.

5. Conclusion

This device presents multiple advantages compared to previously developed "coral-on-a-chip" devices: single well and network configuration, simultaneous dosing of the wells, control over nine inlet/outlets, microsensor retrofitting, top and bottom imaging, minimal flow velocity and shear stress, 3D printable and compact.

The octagon will be an essential part of the feedback loop between experiment biology and bioinformatics focused on reef-building corals. This will facilitate the study of coral protein-protein interactions, calcification, disease & toxicity response, and wound healing from the cellular, polyp and colony level.

Acknowledgements

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References

[1]Shapiro et al., 2015; [2]Gavish et al., 2018; [3]Luo et al., 2020; [4] Pang et al., 2020; [5]Bockel and Rinkevich, 2019; [6]Gibbin et al., 2019; [7]van Treuren et al., 2019; [8]Fick, 1855; [9]Kirby, 2010;

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